Software Requirements Specification for Flexin: subtitle describing software

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Revision History

Date	Version	Notes
Date 1	1.0	Notes
Date 2	1.1	Notes

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Insert your content here.

1.2 Goals of the Project

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2.3 Other Stakeholders

Insert your content here.

2.4 Hands-On Users of the Project

Insert your content here.

2.5 Personas

Insert your content here.

2.6 Priorities Assigned to Users

The project's main priority will be to focus on the beginner - novice users of the application. The goal will be to teach the basis of health/fitness and financial literacy. *Insert your content here.*

2.7 User Participation

Insert your content here.

2.8 Maintenance Users and Service Technicians

Insert your content here.

3 Mandated Constraints

3.1 Solution Constraints

Insert your content here.

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11.1 Ease of Use Requirements

Insert your content here.

11.2 Personalization and Internationalization Requirements

Insert your content here.

11.3 Learning Requirements

ID	Requirement	Specification
UHR-LRN-01	Minimal User Training	**Description:** The system shall enable first-time users to successfully use the application's core functionality without requiring formal training. **Rationale:** The application targets users with varying expertises and should be intuitive for a broad audience. **Fit Criterion:** At least 80% of first-time users shall be able to synchronize fitness data, earn virtual currency, and purchase an investment asset without external assistance.

ID	Requirement	Specification
UHR-LRN-02	Learning Curve of Financial Concepts	**Description:** The system shall present investment concepts in a manner that can be understood by users with little or no prior experience. **Rationale:** One of the system's primary goals is to educate users on financial investing. This means that all users, regardless of their prior financial experience, should be capable of learning and understanding the financial concepts presented. **Fit Criterion:** During user testing, at least 80% of first-time investors shall successfully complete their first virtual investment and view their portfolio's growth after one week without assistance.
UHR-LRN-03	Learning Curve of Fitness Based Concepts	**Description:** The system shall present fitness concepts in a manner that can be understood by users with little or no prior experience. **Rationale:** One of the system's primary goals is to educate users on fitness and health related activities. This means that all users, regardless of their prior fitness experience, should be capable of learning and understanding the fitness concepts presented. **Fit Criterion:** During user testing, at least 80% of first-time participants in a new fitness activity shall successfully accomplish their set goal and view their health metrics after one week without assistance.

ID	Requirement	Specification
UHR-LRN-04	Integrated Educational Assistance	**Description:** The system shall provide contextual learning resources within the application. **Rationale:** In the event that users encounter unfamiliar fitness or financial terminology while using the platform, the system should offer immediate assistance to enhance their understanding so that users can quickly learn and pick up new concepts. **Fit Criterion:** Users shall be able to access explanatory information for all core financial and fitness concepts directly from the screen where the concept appears.
UHR-LRN-05	User Help Documentation	**Description:** The system shall provide readily accessible help documentation describing the application's functionality should any users require it. **Rationale:** Users should be able to resolve common questions without external support. **Fit Criterion:** A help section shall be accessible from the application's settings menu and include explanations of all major features. Support teams should receive zero inquiries about basic system functionality.

11.4 Understandability and Politeness Requirements

Insert your content here.

11.5 Accessibility Requirements

Insert your content here.

12 Performance Requirements

12.1 Speed and Latency Requirements

Insert your content here.

12.2 Safety-Critical Requirements

Insert your content here.

12.3 Precision or Accuracy Requirements

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12.4 Robustness or Fault-Tolerance Requirements

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14 Maintainability and Support Requirements

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14.2 Supportability Requirements

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14.3 Adaptability Requirements

Insert your content here.

15 Security Requirements

15.1 Access Requirements

ID	Requirement	Specification
SEC-ACC-01	Authentication and Sessions	**Description:** Users shall authenticate before accessing personal fitness data, portfolios, or transactions. The React Native client shall authenticate with the Spring API using secure token-based sessions. **Rationale:** Authentication protects personal data and virtual assets from unauthorized access. **Fit Criterion:** All protected API endpoints shall reject missing or invalid access tokens with HTTP 401. Access tokens shall expire within 60 minutes, with refresh or re-authentication required.
SEC-ACC-02	Role-Based Authorization	**Description:** Role-based access control shall restrict users to their own records and limit administrative or evaluator access to authorized functions. **Rationale:** Access restrictions prevent unauthorized modifications and cross-account data exposure. **Fit Criterion:** JUnit security tests shall confirm that attempts by standard users to access or modify another user's records return HTTP 403 in all tested cases.
SEC-ACC-03	Fitness Integration Authorization	**Description:** Flexin shall request only necessary read permissions for supported integrations, Android Health Connect and Garmin. Integration credentials shall be stored using secure, Android Keystore-backed storage. **Rationale:** Secure credential handling protects external accounts and limits access to fitness data. **Fit Criterion:** Integration tokens shall use Android Keystore-backed secure storage. No credentials, tokens, or secrets shall be committed to the project repository.

15.2 Integrity Requirements

ID	Requirement	Specification
SEC-INT-01	Activity Validation	**Description:** The Spring backend shall validate activity metrics and detect duplicate records, overlapping timestamps, and implausible activity values before awarding virtual currency. **Rationale:** Validation prevents activity manipulation and unauthorized currency generation. **Fit Criterion:** Automated tests shall confirm that duplicate or flagged implausible activity records are marked unverified and receive zero virtual currency. Sustained cadence above 250 steps per minute shall be flagged for verification.
SEC-INT-02	Ledger Consistency	**Description:** All virtual currency operations, including rewards, trades, and dividends, shall use ACID-compliant MySQL transactions. **Rationale:** Transactional consistency prevents duplicate rewards, negative balances, and partial updates. **Fit Criterion:** Stress tests with 100 concurrent trade requests against one wallet shall verify that every operation commits consistently or rolls back completely, without partial state changes.

ID	Requirement	Specification
SEC-INT-03	Server-Side Valuation	**Description:** The Spring backend and custom market engine shall determine asset prices, portfolio valuations, and currency calculations. The mobile client shall not control authoritative values. **Rationale:** Server-side calculations prevent client manipulation of prices and balances. **Fit Criterion:** Trade requests attempting to specify unauthorized execution prices shall be rejected with HTTP 400.

15.3 Privacy Requirements

ID	Requirement	Specification
SEC-PRV-01	Health Data Protection	**Description:** Sensitive personal information and fitness telemetry shall be protected through encryption in transit and at rest. **Rationale:** Encryption protects confidential user information against unauthorized disclosure. **Fit Criterion:** Client-server communications shall use TLS 1.3. Sensitive biometric data stored in MySQL shall be encrypted at rest using AES-256.

ID	Requirement	Specification
SEC-PRV-02	Opt-In Community Sharing	**Description:** Leaderboard participation shall be opt-in. Users may share activity streaks and simulated portfolio returns using a chosen alias. **Rationale:** Users should control whether their activity and simulated financial performance are visible to others. **Fit Criterion:** Leaderboard sharing shall be disabled by default. Enabling it shall require explicit consent and disclose which pseudonymous information and statistics will be shared.
SEC-PRV-03	User Data Deletion	**Description:** Users shall be able to request account deletion through the application settings, removing associated personal records. **Rationale:** Users should retain control over their personal information and account history. **Fit Criterion:** Personal health metrics, credentials, and fitness history shall be purged from the active database within 24 hours of a deletion request. Only appropriately anonymized transaction aggregates may be retained for auditing.

15.4 Audit Requirements

ID	Requirement	Specification
SEC-AUD-01	Economic Event Logging	**Description:** Flexin shall maintain an append-only audit ledger for currency rewards, simulated trades, dividends, and market price updates. **Rationale:** Auditable records support troubleshooting, transaction verification, and evaluation of the simulated economy. **Fit Criterion:** Database permissions shall prevent UPDATE and DELETE operations on audit records. Every recorded transaction shall reference a valid source activity or market event, where applicable.
SEC-AUD-02	Security Event Logging	**Description:** The backend shall log failed authentication, token validation failures, duplicate activity submissions, anti-exploit flags, and runtime exceptions. **Rationale:** Security logs help identify suspicious activity and diagnose integration or system failures. **Fit Criterion:** Structured logs shall capture severity, client IP address where available, authenticated user ID where applicable, and exception details for runtime errors.

15.5 Immunity Requirements

ID	Requirement	Specification
SEC-IMM-01	Injection Defense	**Description:** The backend and database layer shall be protected against SQL injection, cross-site scripting (XSS), and command injection. **Rationale:** Injection vulnerabilities could compromise database integrity and expose user information. **Fit Criterion:** Database access shall use parameterized queries or prepared statements. Static analysis and security reviews shall identify zero unresolved high-severity injection vulnerabilities before release.
SEC-IMM-02	Rate Limiting	**Description:** The Spring API shall limit request rates for authentication, trade submission, and activity synchronization endpoints. **Rationale:** Rate limiting reduces brute-force attempts, automated abuse, and service overload. **Fit Criterion:** A configured limit, initially 60 requests per minute per authenticated user or client IP, shall be enforced. Excess requests shall receive HTTP 429. Automated tests shall verify enforcement.

ID	Requirement	Specification
SEC-IMM-03	Client Environment Validation	**Description:** The Android client shall detect supported indicators of rooted or otherwise untrusted environments and restrict sensitive functionality when tampering is detected outside developer builds. **Rationale:** Compromised environments can increase the risk of credential exposure and activity manipulation. **Fit Criterion:** In supported production configurations, detected root or tampering indicators shall trigger restrictions on community leaderboard access and virtual currency minting. Developer builds shall remain available for authorized testing.

16 Cultural Requirements

This section outlines the cultural requirements for the software system. Cultural requirements refer to the expectations and norms of the target user base, which may vary based on geographic location, language, social norms, and other cultural factors. These requirements ensure that the software is culturally sensitive and appropriate for its intended audience.

16.1 Cultural Requirements

ID	Requirement	Specification
CUL-01	Supports Canadian Languages	**Description:** The system shall present all user-facing functionality with the option to choose between English and French. **Rationale:** The application will initially be deployed strictly within Canada, where English and French are the country's official languages. **Fit Criterion:** All user-facing text shall be stored separately from application logic and be replaceable through a localization framework.
CUL-02	Inclusive Financial Education	**Description:** The system shall provide financial education content that does not assume prior investing knowledge. **Rationale:** Users possess varying levels of financial literacy. The system seeks to educate users of all levels of financial literacy - including those who may have limited exposure to investing concepts. **Fit Criterion:** All investment-related terminology presented to users shall include a plain-language explanation accessible from within the application. All financial concepts shall be explained in simple, non-technical language to start, until the system deems the user has gained sufficient experience within the application. During user testing, a survey will be conducted to gather feedback on the clarity and understandibility of the financial education content - at least 80% of users shall express great satisfaction with little to no problems understanding financial concepts.

ID	Requirement	Specification
CUL-03	Inclusive Fitness Participation	**Description:** The system shall accommodate users with varying fitness levels by allowing personalized activity participation. **Rationale:** Users may differ significantly in age, physical ability, health conditions, and exercise experience. The application should promote participation regardless of fitness background. **Fit Criterion:** Users shall be able to participate in their preferred fitness activities while still gaining subsequent rewards.
CUL-04	Culturally Neutral Educational Content	**Description:** The system shall present fitness and financial education content using culturally neutral language and examples (i.e. sample companies in the virtual market). **Rationale:** The Canadian population consists of individuals from diverse cultural, religious, and socioeconomic backgrounds. Educational content should be inclusive and broadly applicable. **Fit Criterion:** Educational content shall not require familiarity with a specific cultural background, religion, ethnicity, or region to be understood.

17 Compliance Requirements

17.1 Legal Requirements

Insert your content here.

17.2 Standards Compliance Requirements

ID	Requirement	Specification
COMP-STD-01	Android Platform Compliance	**Description:** The system shall comply with the Google Play Developer Program Policies. **Rationale:** The application is intended to be deployed on the Google Play Store as to be accessible on Android devices and as a result must conform to platform requirements. **Fit Criterion:** The application shall successfully pass all mandatory checks required for distribution through the Google Play Store.
COMP-STD-02	Accessibility Compliance	**Description:** The system shall comply with WCAG 2.1 Level AA accessibility guidelines for user-facing functionality. **Rationale:** The system should offer a simple and delightful user experience that should be accessible to users with varying abilities (i.e. user's with colour-blindness, hearing impairments, etc.). **Fit Criterion:** All major user workflows shall satisfy WCAG 2.1 Level AA success criteria as evaluated using automated accessibility tools and manual testing.
COMP-STD-03	TypeScript Coding Standards Compliance	**Description:** The frontend codebase shall comply with the Google TypeScript Style Guide. **Rationale:** Consistent coding standards improve maintainability, readability, and long-term project quality. **Fit Criterion:** All pull requests shall pass configured linting and formatting checks before merging into the main branch.

ID	Requirement	Specification
COMP-STD-04	Java Coding Standards Compliance	**Description:** The backend codebase shall comply with the Oracle Java Code Conventions. **Rationale:** Standardized coding practices improve maintainability, code review effectiveness, and software quality. **Fit Criterion:** All pull requests shall pass configured linting and formatting checks before merging into the main branch.
COMP-STD-05	Database Compliance	**Description:** The system shall store data in compliance with the MySQL Reference Manual and the SQL Standard (ISO/IEC 9075). **Rationale:** The project architecture relies upon MySQL as the primary relational database system. **Fit Criterion:** All schema definitions and queries will be tested to execute successfully on the supported MySQL version without compatibility warnings or errors.
COMP-STD-06	Software Testing Standard Compliance	**Description:** The system shall undergo thorough and complete testing in accordance with the IEEE 29119 Software Testing Standard. **Rationale:** Consistent testing practices reduce defects and improve software reliability. The system will comply with this standard to provide a unified, process-driven framework for software testing. **Fit Criterion:** All code merged into the main branch shall pass automated test execution in the CI/CD pipeline based on tests that were developed in accordance with the IEEE 29119 Software Testing Standard.

ID	Requirement	Specification
COMP-STD-07	Information Security Compliance	**Description:** The system shall be designed in accordance with the security principles outlined in ISO/IEC 27001. **Rationale:** The application stores personal, financial-learning, and fitness-related information that must be protected against unauthorized access and disclosure. **Fit Criterion:** Security controls for authentication, access control, data storage, and data transmission shall align with ISO/IEC 27001 recommendations.

18 Open Issues

Insert your content here.

19 Off-the-Shelf Solutions

19.1 Ready-Made Products

Insert your content here.

19.2 Reusable Components

Insert your content here.

19.3 Products That Can Be Copied

Insert your content here.

20 New Problems

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20.2 Effects on the Installed Systems

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20.5 Follow-Up Problems

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21 Tasks

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21.2 Planning of the Development Phases

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22 Migration to the New Product

22.1 Requirements for Migration to the New Product

Insert your content here.

22.2 Data That Has to be Modified or Translated for the New System

Insert your content here.

23 Costs

Insert your content here.

24 User Documentation and Training

24.1 User Documentation Requirements

Insert your content here.

24.2 Training Requirements

Insert your content here.

25 Waiting Room

Insert your content here.

26 Ideas for Solution

Insert your content here.

27 Appendix — Generative AI Use Disclosure

[ChatGPT (version 5) was used to brainstorm ideas for the template for this disclose section. —SS]

27.1 AI Tools Used

[Give the name of the AI tool(s) used and the version. —SS]

27.2 How and Where Used

[Explain what the tool(s) were used for. Examples include: brainstorming, explaining a technical concept, reviewing the document, generating or modifying text, generating or modifying diagrams, identifying errors or omissions or inconsistencies. —SS]

[For each of the uses, identify which parts of the document were affected. —SS]

[You do not need to report every interaction, but you should disclose the uses that materially contributed to your work —SS]

27.3 Verification of AI-Generated Content

[

- Describe how the team checked AI-generated or AI-assisted material before including it in the document.
- In particular, verify factual claims, technical assertions, requirements, calculations, references, and other information for which an AI system may produce incorrect or fabricated results.
- **The team is responsible for the accuracy, completeness, consistency, and appropriateness of the submitted document, regardless of whether material was generated by a human or an AI system.**

—SS]

28 Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?
4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.
5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.
6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?